

Qatar-1b

R-1964

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_131005 · created 2026-08-06T23:10:56+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456570.64915 +/- 0.00014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.15 +/- 0.02
Transit depth (Rp/Rs)^2	2.26 +/- 0.59 [%]
Orbital Inclination (inc)	86.92 +/- 1.54
Airmass coefficient 1 (a1)	1.034 +/- 0.02
Airmass coefficient 2 (a2)	-0.028 +/- 0.064
Scatter in the residuals of the lightcurve fit is	1.93 %
Best Comparison Star	#1 - [370, 139]
Optimal Method	PSF photometry
Transit Duration (day)	0.079 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.15 $\pm$ 0.02 vs 0.1463 (deviation 0.18 σ)
Duration fit vs published	0.079 d vs 0.0692 d (deviation 2.29 σ)
Mid-time O-C	0.3 min
Depth SNR	7.5
Residual scatter	1.93 %

Light curve

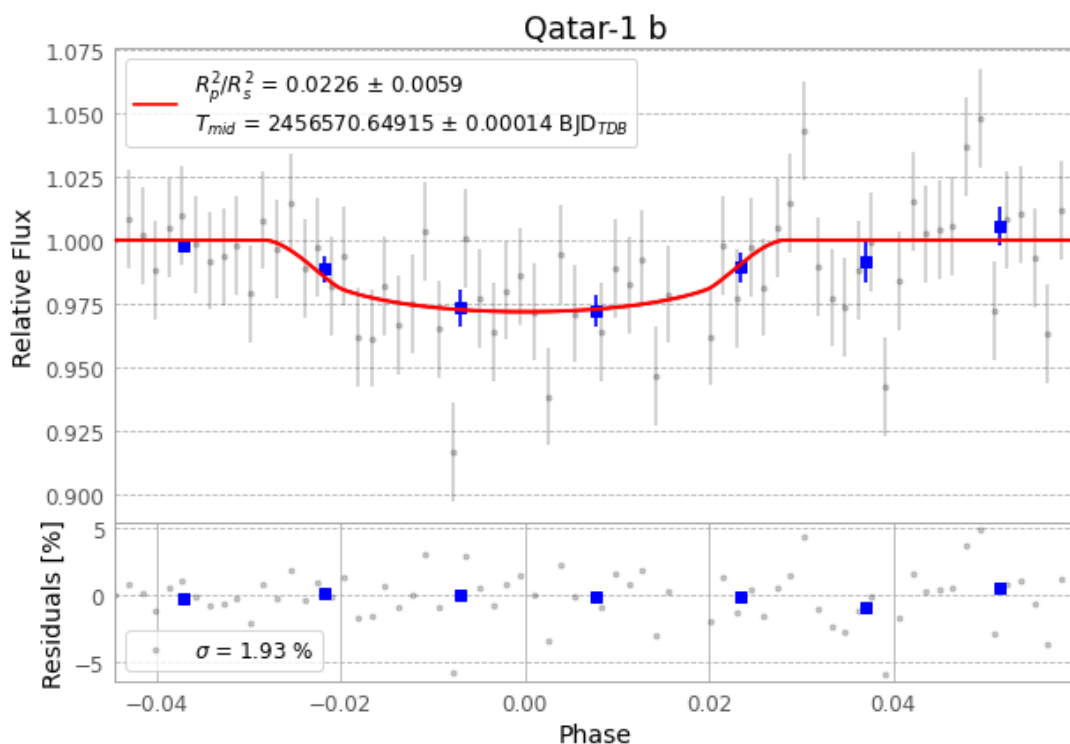


Figure 1 — FinalLightCurve_Qatar-1 b_2013-10-04.png (SHA-256 1f5b6e0e730e287f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [392, 292], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b7c89cdeb7d4921e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

72 FITS frames (47 MB), Qatar-1131005020013.FITS → Qatar-1131005053312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-131005.log (4869379f3283...), FinalLightCurve_Qatar-1 b_2013-10-04.png (1f5b6e0e730e...), FinalLightCurve_Qatar-1 b_2013-10-04.csv (53f4b768faea...)

Compute resources

Wall time 138.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 23:10Z.